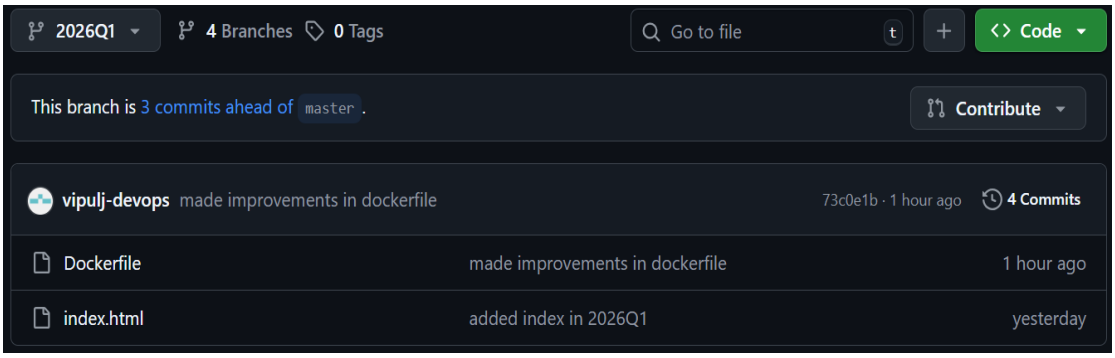
Performed by: Vipul Jadhav

Assignment 3

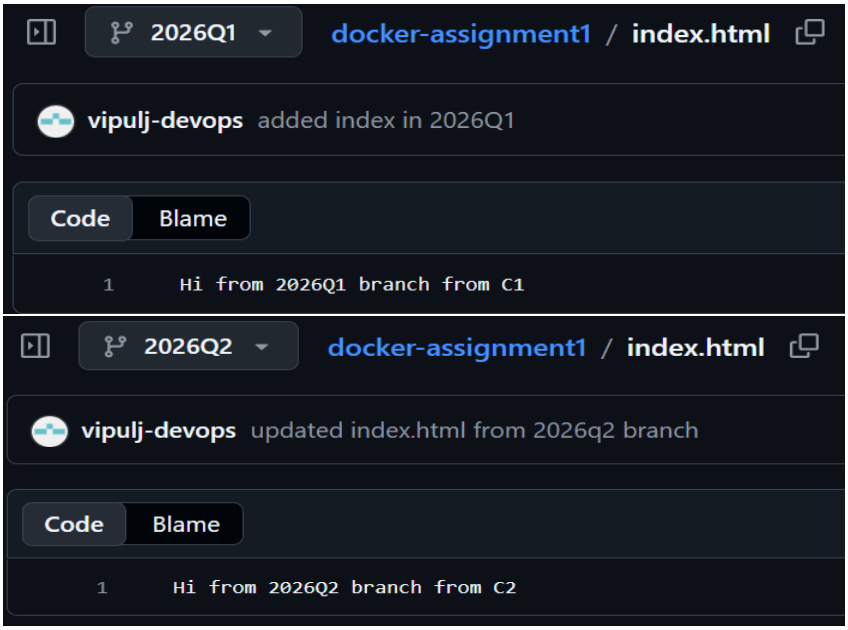
Task: using a Jenkins Declarative Pipeline to build and deploy Docker containers from a Git repository with three branches: 2026Q1, 2026Q2, and 2026Q3, each containing an index.html file. Three containers (c1, c2, and c3) are to be created, where each container hosts the respective branch's index.html file on an Apache HTTP server (httpd). Container c1 should run on port 80, c2 on port 90, and c3 on port 8080 using port binding, with the entire process automated through Jenkins. But now create volume and integrate it in this assignment

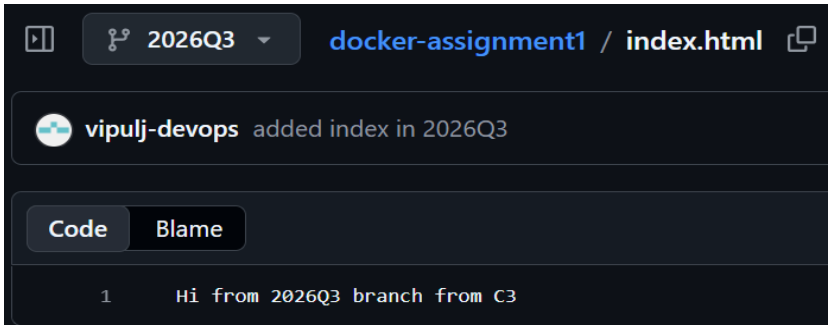
Steps:

1. Created a repository and added index.html and Dockerfile in it

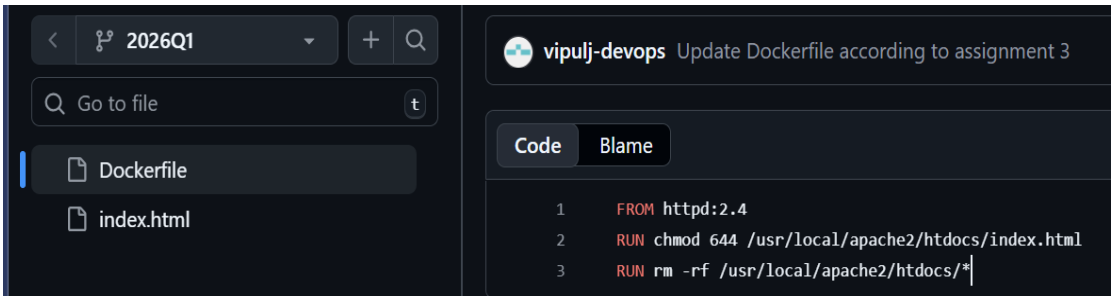


2. Added index.html file in all 3 repositories

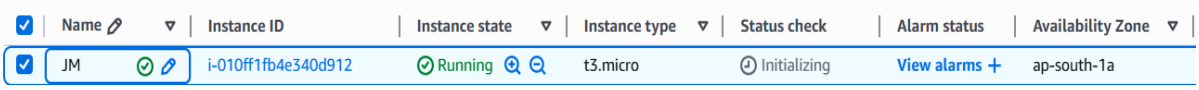




3. Added same Dockerfile in all 3 repositories



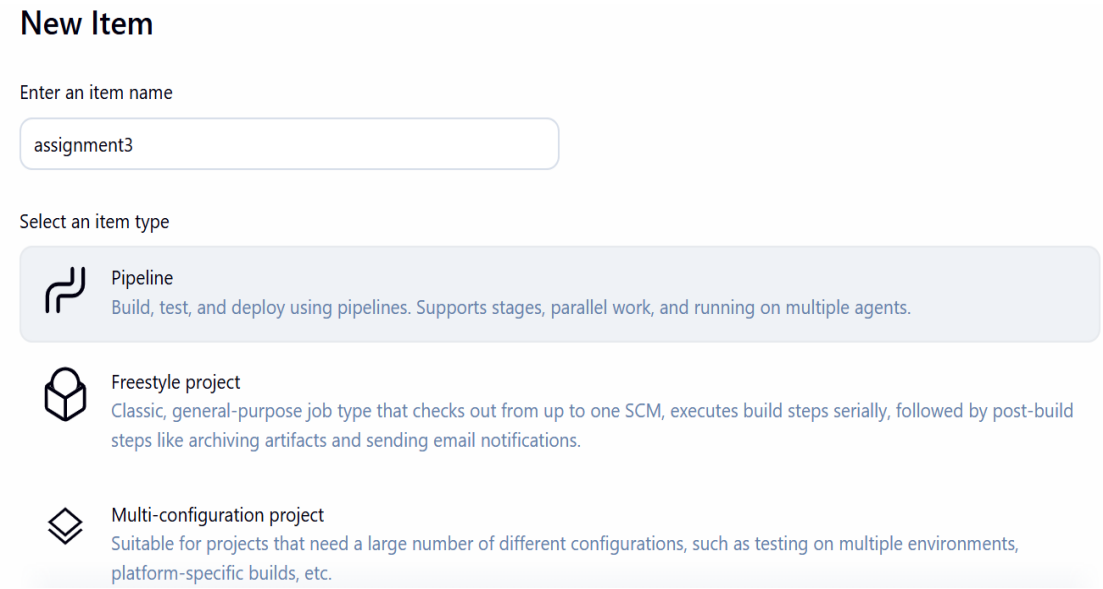
4. Launched Jenkins master container



Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone
JM	i-010ff1fb4e340d912	Running	t3.micro	Initializing	View alarms +	ap-south-1a

5. Installed java-17, docker, tomcat and Jenkins in Jenkins master

6. Created declarative pipeline job in Jenkins master



7. Added Groovy script in the job

Script:

```
pipeline {
  agent any

  stages {

    stage('Clean Workspace') {
      steps {
        deleteDir()
        sh '''
        docker rm -f c1 c2 c3 || true
        docker system prune -a -f || true
        docker volume create deploy || true
        '''
      }
    }

    stage('Build & Run Q1') {
      steps {
        git branch: '2026Q1',
        url: 'https://github.com/vipulj-devops/docker-assignment1.git'

        sh '''
        docker build -t image_q1 .
        cp -f index.html /var/lib/docker/volumes/deploy/_data/
        chmod -R 755 /var/lib/docker/volumes/deploy/_data
        docker run -d -p 80:80 --name c1 -v deploy:/mnt image_q1
        docker exec c1 cp -f /mnt/index.html /usr/local/apache2/htdocs/
        '''
      }
    }

    stage('Build & Run Q2') {
      steps {
        deleteDir()

        git branch: '2026Q2',
        url: 'https://github.com/vipulj-devops/docker-assignment1.git'
```

```
            sh '''
            docker build -t image_q2 .
            cp -f index.html /var/lib/docker/volumes/deploy/_data/
            chmod -R 755 /var/lib/docker/volumes/deploy/_data
            docker run -d -p 90:80 --name c2 -v deploy:/mnt image_q2
            docker exec c2 cp -f /mnt/index.html /usr/local/apache2/htdocs/
            '''
        }
    }

    stage('Build & Run Q3') {
        steps {
            deleteDir()

            git branch: '2026Q3',
            url: 'https://github.com/vipulj-devops/docker-assignment1.git'

            sh '''
            docker build -t image_q3 .
            cp index.html /var/lib/docker/volumes/deploy/_data/
            chmod -R 755 /var/lib/docker/volumes/deploy/_data
            docker run -d -p 8081:80 --name c3 -v deploy:/mnt image_q3
            docker exec c3 cp -f /mnt/index.html /usr/local/apache2/htdocs/
            '''
        }
    }
  }
}
```

Description: This Jenkins declarative pipeline automates the deployment of three Docker containers using different branches of a Git repository. In the first stage, the workspace is cleaned, existing containers are removed, unused Docker resources are cleared, and a Docker volume named deploy is created to store shared files. In each subsequent stage, a specific branch (2026Q1, 2026Q2, 2026Q3) is cloned, and a Docker image is built. The index.html file from each branch is then copied into the Docker volume and appropriate permissions are set to avoid access issues. A container is launched for each branch with the volume mounted at /mnt, and the HTML file is copied into Apache's default directory (/usr/local/apache2/htdocs/) inside the container. Each container runs on a different port (80, 90, and 8081), allowing all three versions of the application to be accessed independently.

8. The job ran successfully

#16 (Mar 31, 2026, 7:03:18 PM)

Started by user vipul

This run spent:

- 25 ms waiting;
- 15 sec build duration;
- 15 sec total from scheduled to completion.

git **Revision**: 256ef044a8b7b18462a060229c0b9e1ac81c15a5
Repository: https://github.com/vipulj-devops/docker-assignment1.git

- refs/remotes/origin/2026Q1

git **Revision**: 3e90b2040ac3893227037d9c41404dc91472740a
Repository: https://github.com/vipulj-devops/docker-assignment1.git

- refs/remotes/origin/2026Q2

git **Revision**: 7ace701d35e255ddbf570945f94a0686383c8983
Repository: https://github.com/vipulj-devops/docker-assignment1.git

- refs/remotes/origin/2026Q3

9. The images and containers got created

Images:

```
[root@ip-172-31-43-203 ec2-user]# docker images
REPOSITORY    TAG       IMAGE ID       CREATED            SIZE
image_q1      latest    74f2721a33e8   20 minutes ago     117MB
image_q2      latest    74f2721a33e8   20 minutes ago     117MB
image_q3      latest    74f2721a33e8   20 minutes ago     117MB
```

Containers:

```
[root@ip-172-31-43-203 ec2-user]# docker ps
CONTAINER ID   IMAGE      COMMAND              CREATED          STATUS          PORTS                                      NAMES
4f849cbb4d07   image_q3   "httpd-foreground"   20 minutes ago   Up 20 minutes   0.0.0.0:8081->80/tcp, :::8081->80/tcp      c3
55bfd45bcec0   image_q2   "httpd-foreground"   20 minutes ago   Up 20 minutes   0.0.0.0:90->80/tcp, :::90->80/tcp          c2
7f92de40836b   image_q1   "httpd-foreground"   21 minutes ago   Up 21 minutes   0.0.0.0:80->80/tcp, :::80->80/tcp          c1
```

10. Index.html got deployed successfully on respective ports

Not secure | 13.126.232.254

Hi from 2026Q1 branch from C1

Not secure | 13.126.232.254:90

Hi from 2026Q2 branch from C2

Not secure | 13.126.232.254:8081

Hi from 2026Q3 branch from C3